Tensor computations

Cohen–Macaulay coordinate rings for every tensor border-rank variety

Difficulty: extreme **Importance:** interesting to specialist

Status: Solved

Rating rationale: An all-format theorem would overcome longstanding obstructions in the local algebra of secant varieties and encompass other major Cohen–Macaulay conjectures. The depth and resolution questions chiefly concern specialists in tensor algebraic geometry. These ratings are historical and refer to the original universal question, which is resolved negatively below. They do not rate the questions that remain open.

Resolution — 4 November 2026

Resolved negatively. Wenqi Zhu (University of Oxford) and Ping Nie (University of Waterloo).

The answer is **no**. For $k = 3$, $(n_1, n_2, n_3) = (3, 4, 5)$ and $r = 5$, the variety $X_5 \subseteq \mathbb{C}^3 \otimes \mathbb{C}^4 \otimes \mathbb{C}^5$ has dimension 50, and its coordinate ring $A_5 = S/I(X_5)$ satisfies

$$\operatorname{depth}(A_5)_{\mathfrak{m}} < 50 = \dim(A_5)_{\mathfrak{m}},$$

so A_5 is not Cohen–Macaulay. This is the case $X_5(3, 4, 5)$ of Corollary 1.2, stated on p. 2 of the 42-page paper *Secant Varieties of Segre Products Need Not Be Cohen–Macaulay* at the immutable revision *3d8b158*, made public on 4 November 2026. Up to the order of the factors, it is the smallest format in the range of Theorem 1.1 (p. 1): for all integers

$$r \geq 5, \qquad a, b \geq 3, \qquad a + b \geq r + 2, \qquad c \geq r,$$

and after any permutation of the three factors, the reduced affine variety $X_r(a, b, c) \subseteq \mathbb{C}^a \otimes \mathbb{C}^b \otimes \mathbb{C}^c$ of tensors of border rank at most r has dimension $D = r(a + b + c - 2)$, fails Serre's condition S_2, is not Cohen–Macaulay, and has depth less than D at its homogeneous maximal ideal. Corollary 1.2 also lists $X_5(6, 6, 6)$, of dimension 80, the varieties $X_r(3, r - 1, r)$ of dimension $2r^2$, and the unit-tensor orbit closures $X_r(r, r, r) = \overline{\mathrm{GL}_r(\mathbb{C})^3 \cdot \Delta_r}$, with $\Delta_r = \sum_{i=1}^{r} e_i \otimes e_i \otimes e_i$, for every $r \geq 5$.

In format $3 \times 4 \times 5$ the witness is the integer tensor $T = \sum_{i=1}^{5} Q_i \otimes f_i$ of equation (4.5) (p. 10), padded with zeros from $3 \times 3 \times 5$. Here $Q_1, \ldots, Q_6$ are the rank-one 3×3 matrices formed from the rows and the negated columns of $M = (j^{i-1})_{1 \leq i,j \leq 3}$, and their only linear relation is $Q_1 + \cdots + Q_6 = 0$ (Lemma 4.1, p. 10). That relation gives a second five-term decomposition (4.7). By Proposition 4.6 (p. 13), T has rank and border rank five and both decompositions have injective addition differentials. Section 2 (pp. 3–8) constructs a smooth birational space of decompositions with an open subset that maps into the smooth locus and has a complement of codimension at least two (Proposition 2.5, p. 7). The two-point conductor obstruction of Proposition 3.3 (p. 9) then shows that the local ring at T is not S_2: the two decompositions give two distinct normalization charts over T, and S_2 would force a conductor divisor that the codimension bound excludes. Section 5 (pp. 14–15) passes to the vertex: if $(A_r)_{\mathfrak{m}}$ were Cohen–Macaulay, a graded free resolution and the Auslander–Buchsbaum formula would make every local ring Cohen–Macaulay.

Comparison with the target. The target asserts Cohen–Macaulayness for every $k \geq 3$, all $n_1, \ldots, n_k \geq 1$ and every $r \geq 1$, so one non-Cohen–Macaulay instance refutes it. The paper's $X_r(a, b, c)$ (equation (1.1), p. 1) is the Zariski closure of the sums of r pure tensors in $\mathbb{C}^a \otimes \mathbb{C}^b \otimes \mathbb{C}^c$ with its reduced structure, so it is the affine cone and contains the zero tensor. Its coordinate ring is $S/I(X_r)$, and $\mathfrak{m}$ is its ideal of positive-degree elements. These are the definitions of the problem statement with $k = 3$ and $(n_1, n_2, n_3) = (a, b, c)$, over the same field $\mathbb{C}$. Since Theorem 1.1 holds after permuting the factors, the order of $(3, 4, 5)$

is immaterial. Corollary 1.3 (p. 2) gives the projective statement: $\sigma_r(\mathrm{Seg}(\mathbb{P}^{a-1}\times\mathbb{P}^{b-1}\times\mathbb{P}^{c-1}))$ has a non-Cohen–Macaulay local ring at the class of the witness, so it is neither locally nor arithmetically Cohen–Macaulay. Section 1.1 (p. 2) notes that Conjecture 1.1 of reference 1 and Conjecture 3.32 of reference 2 concern the ordinary reduced secant varieties, with neither a conciseness condition nor a restriction to balanced formats, and that Theorem 1.1 refutes them with the same field, reduced structure and projective embedding.

The unit-tensor orbit closures $X_r(r,r,r)$ were already known to be nonnormal for every $r \geq 5$, by Bürgisser and Ikenmeyer (Corollary 7.2(2) of arXiv:1011.1350v1 and Corollary 5.26 of arXiv:1511.02927v2), as cited in Section 1.2 (pp. 2–3). Nonnormality does not imply failure of S_2 or of Cohen–Macaulayness; the paper's example is the Cohen–Macaulay, nonnormal domain $\mathbb{C}[t^2,t^3]$. As the paper states, Theorem 1.1 strengthens the known nonnormality of these orbit closures to failure of S_2 and of Cohen–Macaulayness, and the proof does not use the nonnormality results. In reference 2, implications run from the universal conjecture to Cohen–Macaulayness assertions for surface Quot schemes and smoothable Hilbert scheme components. Section 6.4 (pp. 18–19) notes that its failure gives no counterexample to these assertions or to Haiman's isospectral Hilbert scheme conjecture; Propositions 6.3 and 6.4 (pp. 17–18) show that the mechanism cannot occur on invertible-slice charts and that the witness lifts to smooth points of the concise secant. The counterexample refutes only the universal assertion. It does not decide Cohen–Macaulayness in formats and ranks outside Theorem 1.1, apart from the extensions below.

Extensions. In the other direction, Proposition 6.2 (p. 16) shows that $X_5(3,3,c)$ is Cohen–Macaulay for every $c \geq 1$: it is the whole space for $c \leq 5$ and the variety of $9\times c$ matrices of rank at most five for $c \geq 5$. Section 7 (pp. 19–40) describes the normalization near explicit witnesses. Corollary 7.3 (p. 20) gives exactly six reduced, smooth normalization points over the rank-five witness for all $a,b \geq 3$, $a+b \geq 7$ and $c \geq 5$, and Theorems 7.6, 7.8 and 7.9 (pp. 22–26) give exact smooth fibres of six, seven and $r+1$ points over higher-rank witnesses. Theorem 7.10 (p. 27) presents the completed local ring at the rank-five witness, with dimension $5(a+b+c-2)$ and depth $3a+3b+5c+3$; in format $3\times4\times5$ these are 50 and 49 (Remark 7.11, p. 30). Theorems 7.12–7.14 (pp. 31–32) give the normalization fibres and the local models of their boundary points for one-sided extensions and for two specified mixed extensions. These results concern the displayed witnesses.

What remains open. The paper determines neither the depth at the homogeneous vertex nor, for $r \geq 6$, the depth at the witness (p. 2; Remark 7.11, p. 30). It makes no minimal-format assertion (pp. 3 and 17), and formats outside the theorem, such as $X_5(3,4,4)$ and $X_5(4,4,4)$, in which no factor has dimension at least five, are not decided. Theorem 6.1 (p. 15) shows that no three-factor tensor has two distinct immersive r-term decompositions when $r \leq 4$, so the mechanism first occurs at rank five; this does not decide Cohen–Macaulayness for $r \leq 4$ (p. 16). The paper gives no classification of the Cohen–Macaulay border-rank varieties, none in order $k \geq 4$, and no Cohen–Macaulay or rational-singularity assertion about their normalizations (pp. 3 and 19).

Proof and verification scope. Theorem 1.1, its corollaries and the extensions are proved in the written paper. A Lean formalization at the same revision proves exactly two finite certificate layers, in two pinned projects using only the standard axioms. *TR25.finiteCertificate*, recorded in its Lean guide, covers the $3\times3\times5$ integer witness and its zero padding to $6\times6\times6$: both decompositions with nonzero, pairwise nonproportional summands and different directions, third flattening rank five, the six rational determinant values in (4.11) and (4.12), and coordinate Jacobian ranks 45 and 80. *TR25Family.finiteCertificate*, recorded in its Lean guide, covers the witness in format $3\times4\times5$: both decompositions, flattening ranks $3,3,5$, the pair-product and third-factor ranks, the six determinant values and both coordinate Jacobian ranks 50. Neither formalizes border rank, the dimension of the secant variety, the geometric tangent spaces, normalization, the conductor argument, S_2, Cohen–Macaulayness, any depth statement (vertex or local) or the family for all ranks and formats. No numbered result of the paper is a Lean theorem in full, and no Lean theorem states the failure of Cohen–Macaulayness, so this is not a Lean verification of the target. The verification record also describes exact checks of the witness calculations, of the finite cores of the Section 7 results and of 31 pairs of family decompositions, which certify finite calculations and do not establish the theorem.

Review. Solved by independent informal AI-agent audits: two reports, by different referees, on each written result (the $6 \times 6 \times 6$ case, the family theorem, the normalization fibres, the higher-rank fibres, the completed local ring and the one-sided and mixed boundary models), reports on the statement and on the proof of each Lean certificate, and reports on three earlier drafts of the paper; none requested a mathematical correction. The review record contains the audit reports and records their scope and its independence caveats. The paper discloses its use of AI tools. The catalogue did not rerun the Lean or exact checks.

The retained status check below predates this resolution.

Problem statement

Fix integers $k \geq 3$, $n_1, \ldots, n_k \geq 1$, and $r \geq 1$. In $V = \mathbb{C}^{n_1} \otimes \cdots \otimes \mathbb{C}^{n_k}$, let

$$X_r = \overline{\left\{ \sum_{\ell=1}^{r} v_{1,\ell} \otimes \cdots \otimes v_{k,\ell} : v_{i,\ell} \in \mathbb{C}^{n_i} \right\}}^{\mathrm{Zar}}.$$

Let $S = \mathbb{C}[V]$ be the polynomial ring in the tensor entries, let $I(X_r)$ be its vanishing ideal, and let $A_r = S/I(X_r)$.

Is A_r Cohen–Macaulay for every such format and rank threshold?

Explicitly, if $\mathfrak{m}$ is the ideal of positive-degree elements of A_r, the requested equality is

$$\operatorname{depth}(A_r)_{\mathfrak{m}} = \dim(A_r)_{\mathfrak{m}}.$$

Depth is the maximum length of a regular sequence in the maximal ideal: each successive element is a non-zero-divisor after quotienting by its predecessors. Dimension is Krull dimension. This is Oeding's conjecture that all secant varieties of Segre products are arithmetically Cohen–Macaulay. The zero tensor is retained by using the affine cone.

Why it matters

These are the algebraic sets underlying low-border-rank tensor approximation. Cohen–Macaulayness constrains their equations and free resolutions, supporting the determination of generators and the analysis of degeneracies. The conjecture does not assert that exact low-rank tensor approximation is computationally easy or always attained.

References and status check

1. L. Oeding, *Are all Secant Varieties of Segre Products Arithmetically Cohen-Macaulay?*, Preprint (2016), v2, Conjecture 1.1, p.1; §2 collects established cases.
2. J. Jagiełła and J. Jelisiejew, *Unrestrictions and concise secant varieties*, 2026 preprint, v2, Conjecture 3.32 and its surrounding discussion; §1.5.3 explains a connection to the Salmon ideal problem.

Status check — 2026-09-10

Checked Oeding v2, including Theorems 3.1–3.2 and Proposition 3.5, and Jagiełła–Jelisiejew v2, Conjecture 3.32, with targeted later-resolution searches. Oeding proves substantive tensor cases, including the fourth secant of $\mathbb{P}^2 \times \mathbb{P}^2 \times \mathbb{P}^n$ for $n \geq 3$, rather than only the matrix or rank-one cases. The June 2026 source still treats the all-format conjecture as open. Inheritance results require extra hypotheses and do not establish it universally. No full proof or counterexample was located.